

Advanced (Carolina Reaper)

Q1. Factor the expression $x^3 + 8$

- (A) $(x + 2)(x^2 - 2x + 4)$
- (B) $(x - 2)(x^2 + 2x + 4)$
- (C) $(x + 2)(x^2 + 2x + 4)$
- (D) $x(x^2 + 8)$
- (E) $x^2(x + 8)$

Q2. Factor the expression $27x^3 - 8y^3$

- (A) $(3x - 2y)(9x^2 + 6xy + 4y^2)$
- (B) $(3x - 2y)(9x^2 - 6xy + 4y^2)$
- (C) $(3x + 2y)(9x^2 - 6xy + 4y^2)$
- (D) $(3x + 2y)(9x^2 + 6xy + 4y^2)$
- (E) $(3x - 2y)(3x^2 - 2xy + y^2)$

Q3. What is the factored form of $x^3 - 64$?

- (A) $(x - 4)(x^2 + 4x + 16)$
- (B) $(x + 4)(x^2 - 4x + 16)$
- (C) $(x - 4)(x^2 - 4x + 16)$
- (D) $(x + 4)(x^2 + 4x + 16)$
- (E) $(x - 4)(x^2 + 4x - 16)$

Q4. Factor the expression $8x^3 + 27$

- (A) $(2x + 3)(4x^2 - 6x + 9)$
- (B) $(2x - 3)(4x^2 + 6x + 9)$
- (C) $(2x + 3)(4x^2 + 6x + 9)$
- (D) $(2x - 3)(4x^2 - 6x + 9)$
- (E) $2(4x^3 + 27)$

Q5. What is the factored form of $x^3 + 125$?

- (A) $(x + 5)(x^2 - 5x + 25)$
- (B) $(x - 5)(x^2 + 5x + 25)$
- (C) $(x + 5)(x^2 + 5x + 25)$
- (D) $(x - 5)(x^2 - 5x + 25)$
- (E) $x(x^2 + 125)$

Q6. Factor the expression $64x^3 - 1$

- (A) $((4x - 1)(16x^2 + 4x + 1))$
- (B) $((4x + 1)(16x^2 - 4x + 1))$
- (C) $((2x + 1)(32x^2 - 2x + 1))$
- (D) $((2x - 1)(32x^2 + 2x + 1))$
- (E) $(8x - 1)(8x^2 + 1)$

Q7. What is the factored form of $x^3 - 1000$?

- (A) $(x - 10)(x^2 + 10x + 100)$
- (B) $(x + 10)(x^2 - 10x + 100)$
- (C) $(x - 10)(x^2 - 10x + 100)$
- (D) $(x + 10)(x^2 + 10x + 100)$
- (E) $x(x^2 - 1000)$

Q8. Factor the expression $125x^3 + 8y^3$

- (A) $(5x + 2y)(25x^2 - 10xy + 4y^2)$
- (B) $(5x + 2y)(25x^2 + 10xy + 4y^2)$
- (C) $(5x - 2y)(25x^2 + 10xy + 4y^2)$
- (D) $(5x - 2y)(25x^2 - 10xy + 4y^2)$
- (E) $(5x + 2y)(25x^2 - 10xy - 4y^2)$

Open Ended Questions (FRQ)

Q9. Factor the expression $x^3 + 343$ and explain your reasoning.

Q10. Factor the expression $64x^3 - 125y^3$ and show your work.

Chef's Tips

- Make sure students show all work and explain their reasoning for FRQ questions
- Remind students to check their work carefully before submitting
- Consider providing a formula sheet with the sum and difference of cubes formulas